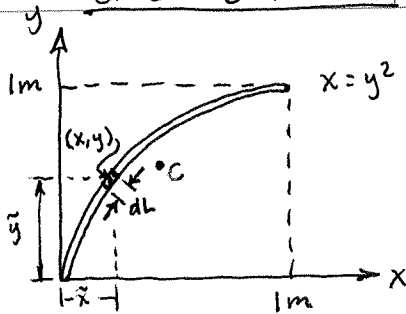
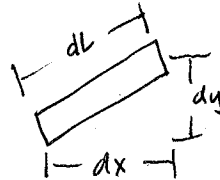


LINE CENTROIDS

Hibbeler Example 9-1



Looking at the element:



$$dL = \sqrt{dx^2 + dy^2}$$

There are several choices of ways to work this problem. Some are easier. All lead, ultimately, to the same result.

$$dL = \sqrt{\left(\frac{dx}{dy}\right)^2 + 1} dy$$

$$dL = \sqrt{dx^2 + dy^2}$$

Factor out $\sqrt{dy^2}$

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$$dL = \sqrt{\left(\frac{dx}{dy}\right)^2 + 1} dy$$

$$dL = \sqrt{\left(\frac{dy}{dx}\right)^2 + 1} dx$$

A.

(Book's way)

$$x = y^2$$

$$dx = 2y dy$$

$$\frac{dx}{dy} = 2y$$

$$dL = \sqrt{4y^2 + 1} dy$$

B.

$$y = \sqrt{x} = x^{1/2}$$

$$dy = \frac{1}{2} x^{-1/2} dx$$

$$\frac{dy}{dx} = \frac{1}{2} x^{-1/2}$$

$$dL = \sqrt{\left(\frac{1}{2} x^{-1/2}\right)^2 + 1} \frac{1}{2} x^{-1/2} dx$$

$$= \left(\sqrt{\frac{1}{4x} + 1} \right) \frac{1}{2} x^{-1/2} dx$$

$$= \sqrt{1 + \frac{1}{4x}} dx$$

$$dL = \sqrt{4y^2 + 1} dy$$

Since $x = y^2$

$$\text{and } dy = \frac{1}{2} x^{-1/2} dx$$

C.

$$x = y^2$$

$$dx = 2y dy$$

$$\frac{dx}{dy} = \frac{1}{2y}$$

$$dL = \sqrt{\left(\frac{1}{2y}\right)^2 + 1} 2y dy$$

$$dL = \sqrt{1 + 4y^2} dy$$

Since $x = y^2$

D.

$$y = \sqrt{x}$$

$$dy = \frac{1}{2} x^{-1/2} dx$$

$$\frac{dy}{dx} = \frac{1}{2} x^{-1/2}$$

$$dx = 2x^{1/2} dy$$

$$dL = \sqrt{\left(\frac{1}{2} x^{-1/2}\right)^2 + 1} dx$$

$$dL = \sqrt{\frac{1}{4x} + 1} (2x^{1/2}) dy$$

$$dL = \sqrt{1 + 4x} dy$$

$$dL = \sqrt{1 + 4y^2} dy$$

$$\bar{x} = \frac{\int_L \bar{x} dL}{\int_L dL} = \frac{\int_0^1 x \sqrt{4y^2 + 1} dy}{\int_0^1 \sqrt{4y^2 + 1} dy} = \frac{\int_0^1 y^2 \sqrt{4y^2 + 1} dy}{\int_0^1 \sqrt{4y^2 + 1} dy} = \frac{.6063}{1.479} = 0.410 \text{ m} = \bar{x}$$

$$\bar{y} = \frac{\int_L \bar{y} dL}{\int_L dL} = \frac{\int_0^1 y \sqrt{4y^2 + 1} dy}{\int_0^1 \sqrt{4y^2 + 1} dy} = \frac{.8484}{1.479} = .5736 \text{ m} = \bar{y}$$

Use Integration Tables in Appendix A, pp. 508, 509

For example, $L = \int dL = \int_0^1 \sqrt{4y^2 + 1} dy = \int_0^1 2\sqrt{y^2 + 1/4} dy$

$a = 1/2$

$$\left\{ \int \sqrt{x^2 \pm a^2} dx = \frac{1}{2} \left[x \sqrt{x^2 \pm a^2} \pm a^2 \ln(x + \sqrt{x^2 \pm a^2}) \right] + C \right\}$$

$$\int_0^1 2\sqrt{y^2 + (1/2)^2} dy = 2 \left(\frac{1}{2} \right) \left[x \sqrt{x^2 + 1/4} + \frac{1}{4} \ln(x + \sqrt{x^2 + 1/4}) \right]_0^1 = 1 \cdot \left[\sqrt{1.25} + \frac{1}{4} \ln(1 + \sqrt{1.25}) - 0 - \frac{1}{4} \ln(1/2) \right]$$

$$L = 1.479 \text{ m}$$

Hibbeler Ex Prob 9-1 (Line Centroid)

Eval these integrals on TI-89:

$$\bar{x} = \frac{\int_0^1 y^2 \sqrt{4y^2 + 1} dy}{\int_0^1 \sqrt{4y^2 + 1} dy} = \frac{0.6063}{1.479} = \boxed{0.410 \text{ m} = \bar{x}}$$

Denominator = length of line

$$\boxed{\text{2ND}} + \boxed{7} \Rightarrow \int (\quad \quad \quad \boxed{\text{2ND}} + \boxed{x} \Rightarrow \sqrt{($$

TI-89: $\int (\sqrt{4 * y^2 + 1}, y, 0, 1) = 1.479 \text{ m}$

$\bar{x}$ Numerator: $\int (y^2 * \sqrt{4 * y^2 + 1}, y, 0, 1) = 0.6063$

$\bar{y}$ Numerator = $\int_0^1 y \sqrt{4y^2 + 1} dy$

TI-89: $\int (y * \sqrt{4 * y^2 + 1}, y, 0, 1) = 0.8484$

$$\bar{y} = \frac{0.8484}{1.479} = \boxed{0.5736 \text{ m} = \bar{y}}$$